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Practice Learning Check - Learning the Feature Vector

Instructions

Review Learning Objectives

1. How can we create random features when we are unsure what to use?

1 / 1 point

☒ We can apply a random matrix to the features to generate random linear combinations of the input elements.

Ready to review what you've learned before starting the assignment? I'm here to help.

Help me practice

Let's chat

☐ We replace the input elements with random variables.

☒ Correct

The paper by Rahimi and Recht (2008) proposed the idea that for shift-invariant kernels (which have the property $k(x, x') = k(x - x')$, one may use a matrix with random elements as our SL.

Assignment details

Submitted

Attempts

Oct 5, 5:23 PM EDT

4 left (5 attempts every 24 hours)

Try again

2. Is the following statement true or false?

1 / 1 point

One benefit of using random vectors is that it lets us create a lot of features quickly.

☒ True

☐ False

Your grade

To pass you need at least 80%. We keep your highest score.

100%

View submission

See feedback

☒ Correct

Correct, but recall — using a random matrix is a cheap trick; it lets us create a lot of features quickly without worrying about their quality.

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3. What's the benefit of using a cosine function for these types of problems?

1 / 1 point

☒ It makes the learning function non-linear.

☐ It makes the learning function linear.

☒ Correct

The function $\sigma(\cdot)$ is a nonlinear function of its argument and acts on all elements of the argument element-wise